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Artificial intelligence image-based prediction models in IBD exhibit high risk of bias: A systematic review

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ABSTRACT

Background: There has been an increase in the development of both machine learning (ML) and deep learning (DL) prediction models in Inflammatory Bowel Disease. We aim in this systematic review to assess the methodological quality and risk of bias of ML and DL IBD image-based prediction studies.
Methods: We searched three databases, PubMed, Scopus and Embase, to identify ML and DL diagnostic or prognostic predictive models using imaging data in IBD, to Dec 31, 2022. We restricted our search to include studies that primarily used conventional imaging data, were undertaken in human participants, and published in English. Two reviewers independently reviewed the abstracts. The methodological quality of the studies was determined, and risk of bias evaluated using the prediction risk of bias assessment tool (PROBAST).
Results: Forty studies were included, thirty-nine developed diagnostic models. Seven studies utilized ML approaches, six were retrospective and none used multicenter data for model development. Thirty-three studies utilized DL approaches, ten were prospective, and twelve multicenter studies. Overall, all studies demonstrated high risk of bias. ML studies were evaluated in 4 domains all rated as high risk of bias: participants (6/7), predictors (1/7), outcome (3/7), and analysis (7/7), and DL studies evaluated in 3 domains: participants (24/33), outcome (10/33), and analysis (18/33). The majority of image-based studies used colonoscopy images.
Conclusion: The risk of bias was high in AI IBD image-based prediction models, owing to insufficient sample size, unreported missingness and lack of an external validation cohort. Models with a high risk of bias are unlikely to be generalizable and suitable for clinical implementation.

1. Introduction

At present, there is an increasing interest in harnessing the potential of Artificial Intelligence (AI) algorithms across healthcare. Inflammatory Bowel Disease (IBD) is no exception, with increasing availability of big data and advancements in computational approaches the potential opportunities are vast [1]. The diagnosis and effective management of IBD requires multi-modal imaging data (endoscopic, histological and magnetic resonance imaging (MRI)), which provides a unique opportunity to leverage AI and learn complex patterns from fine imaging, to derive predictive algorithms to improve clinical care. However, very few algorithms in healthcare have been deployed and implemented in clinical practice [2], with AI based models potentially further contributing to research waste [3]. Thus, it is critical to study the current limitations of such models in IBD.

The terms artificial intelligence (AI) and machine learning (ML) are often used interchangeably, despite notable differences. AI is a broad concept, referring to computer programs that can mimic human cognitive function [4]. Machine learning is an application of AI, a discipline at the intersection of computer science and statistics, that aims to identify and learn from patterns in data [4]. Machine learning models can either be supervised or unsupervised. Supervised learning requires both the input (predictors) and outcome data to be labeled and is typically undertaken for classification or regression problems. Unsupervised learning uses unlabelled data to identify patterns, commonly for clustering, representation learning and density estimation tasks. Unsupervised learning is useful in both exploratory analysis and dimension reduction. Deep learning (DL) models emulate the neuronal architecture

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of the brain. DL uses complex artificial neural networks (multiple layers of processing) to automatically detect and transform higher level features from the input data, into global learning of the classification task [5]. A great deal of research to date in gastroenterology [6] has focused on computer vision (CV), an interdisciplinary scientific field that deals with how computers can obtain meaningful data from visual inputs (e.g. digital images or videos). Characteristically CV, employs a type of artificial neural network called convolutional neural network (CNN) models. Table one summarizes the popular machine learning and deep learning algorithms, and example study pipelines are shown in Fig. 1. The terms "model" and "algorithm" have been used interchangeably in this review.

We undertook a systematic review to assess the methodological quality and risk of bias of studies on machine learning and deep learning for image-based prediction models in inflammatory bowel disease.

2. Methods

The systematic review followed PRIMSA [7] (preferred reporting items for systematic reviews and meta-analyses). We appraised the risk of bias using the prediction model risk of bias assessment (PROBAST) [8]. We adapted PROBAST to be applicable for machine (ML) and deep learning (DL) methodological approaches (supplementary S1).

2.1. Literature search

We searched three databases, PubMed, Scopus, and Embase, for ML and DL prediction models utilizing imaging data, for diagnosis or prognosis in IBD, published in English and to Dec 31, 2022. An electronic search was performed querying for diseases types (inflammatory bowel disease (IBD), ulcerative colitis, Crohn's disease), image types (endoscopy, capsule endoscopy, colonoscopy, pathology, histopathology, medical imaging, whole-slide image, wireless capsule endoscopy) in the title, and AI methods in either abstract or title (machine learning/machine intelligence, deep learning, artificial intelligence, neural language processing, neural networks (computer), computational intelligence, support vector, random forest. The search query utilized both MeSH terms and specific search terms. Details of the search query are provided in supplementary S2.

2.2. Inclusion and exclusion criteria

Eligible publications included ML or DL predictive models that were either (1) diagnostic; with the aim to classify IBD disease type or severity, or (2) prognostic; to predict treatment response or a clinical outcome. We further, restricted our search to include studies that primarily used conventional imaging data (e.g., colonoscopy, video capsule endoscopy, pathology, etc.), studies undertaken in human participants and written in English. We excluded studies if only regression-type techniques were employed (linear regression, logistic regression, lasso regression, ridge regression, or elastic net). Systematic reviews, methodological articles, and conference abstracts were also excluded.

2.3. Screening process

Two independent reviewers (XL, JR) screened titles and abstracts. A third reviewer helped to resolve disagreements (JD). After potentially eligible studies had been selected, one researcher (XL) retrieved full-text articles for eligibility, and two researchers independently (XL, JD) extracted data for assessment of risk of bias.

2.4. Data extraction & analysis

We developed two data extraction forms. We collected data on the following study type, patient, and model characteristics with a standardized study report form: (1) study type including study year, IBD disease type, task (diagnosis or prognosis); (2) patient characteristics including number of patients in study, number of imaging exams; (3) model characteristics included type of ML/DL model, cross validation or external validation, model performance metrics (accuracy, specificity, sensitivity, area under the receiver operator curve (AUROC)). We utilized the PROBAST tool, which consists of 21 signaling questions in four domains (participants, predictors, result, and analysis) to appraise the ML/DL models. We omitted signaling questions that were deemed not relevant for appraising ML/DL models [9,10]. Specifically, for DL models, we did not evaluate predictor domain questions, because the predictor was the individual pixels of the image [9]. The risk of bias signaling questions and rationale for exclusion are described in detail in supplementary S2.

We evaluated every signaling question and then gave each study category an overall grade. If all signaling questions in each domain are answered as "yes", PROBAST deems the domain to be low risk. If one signaling question was answered as "no" or "insufficient/unclear information", PROBAST recommends rating the domain as probable high risk or unclear, respectively. If at least one domain in the study is deemed to be high risk, the overall judgment for ML or DL study will be

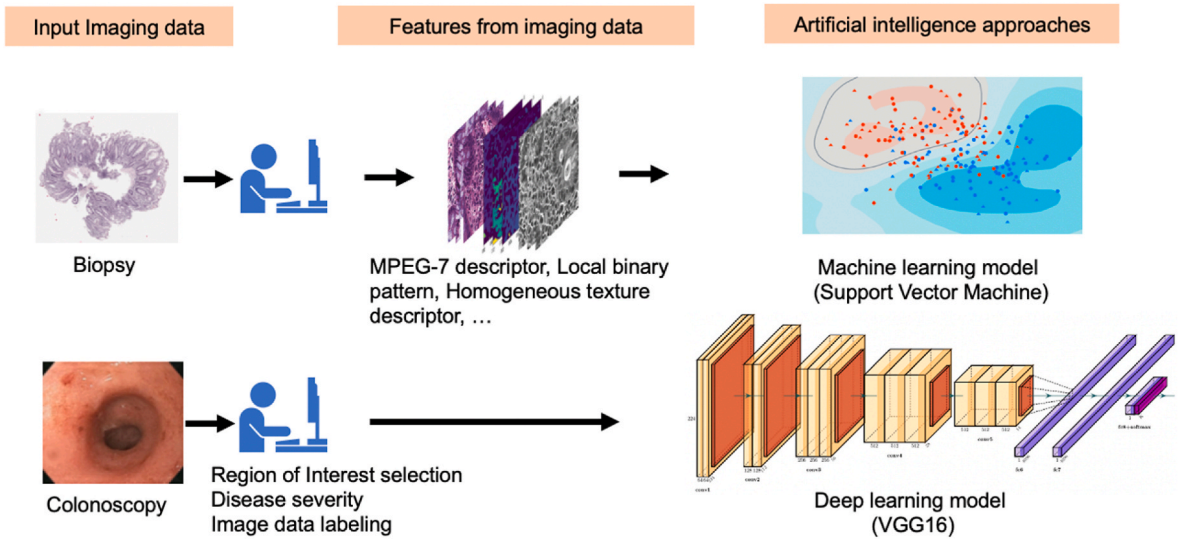

Fig. 1. IMAGE-BASED study design pipeline..

assigned as high risk of bias [8]. We categorized the model studies of these publications into either ML or DL model.

3. Results

Our systematic search identified 1122 articles with no duplicate studies. A total of 462 underwent full-text screening, and 421 were excluded. Forty articles were included in the final appraisal (Fig. 2).

The majority of studies were retrospective, with only eleven (10 DL and 1 ML) studies undertaken prospectively. Twelve studies were multicenter, and four studies [11–14] used two public dataset (Hyper-Kvasir [15], and CrohnIPI [16]). Three studies did not provide any information on the data source for model development [17–19]. Fifteen studies (5 ML studies, 10 DL studies) developed models using data from less than 100 patients (min 2 to max 3082), and three DL studies did not provide the number of participants. Majority of the predictive models were diagnostic (only one model was prognostic [20]), and were either characterizing inflammation from normal or predicting the degree of inflammation (e.g., Endoscopic Mayo Score) (see Table 1).

3.1. Overview of image data driven machine learning models

Seven studies utilized ML approaches, and six were diagnostic and retrospective (details of the selected studies are presented in Table 2). The one prognostic study was a prospective cohort study. The study derived a model using input from colonoscopy videos to predict clinical relapse (defined as partial Mayo endoscopic score>2) in UC patients, during a year of follow-up [20]. Five studies used datasets with less than 100 patients, and one study derived a model using data from only two patients [21]. Maeda et al. used imaging data from 145 patients from a clinical trial [20]. Only one study provided details of sex distribution, and no studies provided information on race or ethnicity. Three of the seven studies identified inflammatory lesions in CD, three studies used still images (frames) from wireless capsule endoscopy video [21–23]. Kumar et al. [23] employed a ML model to detect lesions and normal endoscopic frames and graded the frames from mild to severe. Two of the studies distinguished UC/CD patients from healthy control subjects by using computed tomography (CT) and multi-slice CT [22,24]. Li T et al. [24] utilized CT images to characterize active CD.

All studies used handcrafted feature methods to extract features from the frames/images. Handcrafted features are mathematically defined using feature engineering techniques, encoding repeated patterns and distinct image characteristics such as texture, shape, color, morphology and spatial orientations in a given image. The unstructured imaging data (pixels within the region of interest) are converted into structured data that can be used for model training. Color description tools included representative colors (DominantColor), color distribution (ScalableColor) and spatial distribution of colors (ColorStructure) [25]. Texture features included Homogeneous Texture Descriptor (HTD) and the Edge Histogram Descriptor (EHD), and shape features included RegionShape, ContourShape, and Shape3D [26].

In all the studies the features were used to train support vector machine (SVM) classifiers. Two studies used cross-validation and none of the studies evaluated their model using an external validation dataset.

3.2. Assessment of risk of bias in machine learning models

Six of seven studies were rated high risk of bias for participants domains. Three studies provided details of their inclusion and exclusion criteria for data selection [20,24,27] (signaling Q1.2). Six of seven studies were rated low risk of bias for the predictor domain. Seven studies used handcrafted features from imaging data, and these were assessed in a similar way for all participants included in the study. One study used both handcrafted features and clinical features to derive a ML model to discriminate between UC and CD[27]. All handcrafted features used in ML studies were assessed without knowledge of the outcome

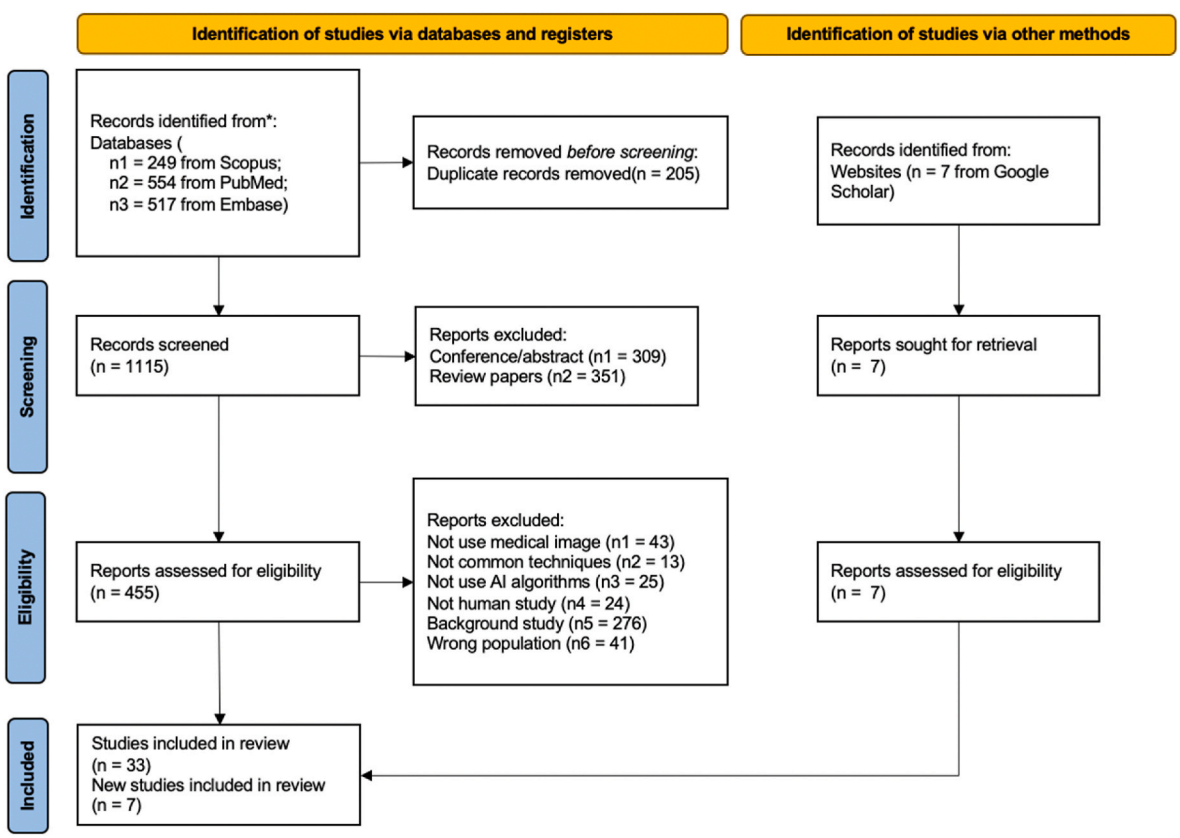

Fig. 2. Prisma flowchart

Table 1
Commonly used machine and deep learning models in the imaging DOMAIN.

Machine learning		
	Models	Explanation
unsupervised	Principal Component Analysis (PCA)	A model that computes the principal component to reduce the dimension of the input data without losing a significant amount of information [83].
supervised	K-nearest Neighbors (KNN)	A non-parametric classification method that categorizes inputs by the similarity between data points [84].
supervised	Logistic Regression (LR)	A model uses coefficients between data points to estimate the probabilities of classes [85].
supervised	Naive Bayes	A model based on Bayes theorem.
supervised	Random Forest (RF)	An ensemble of decision trees and feature selection that predicts classes of data points [86]. RF has the advantage of dealing with high-dimensional feature inputs and complex data structures.
supervised	Support Vector Machine (SVM)	A model that defines a hyperplane or a set-of-hyperplane depending on the number of classes; the hyperplane can be a straight line or represented by a kernel [87].
supervised	Extreme Gradient Boosting (XGBoost)	A regularized form of Gradient Boosting model.
Deep learning models (Convolutional Neural Network)		
Models		**Explanation**
VGG16, VGG19		A CNN model proposed by Simonyan and Zisserman [88]. This model consists of 13 convolutional layers, five max-pooling layers, three dense layers, and one softmax layer. The difference between VGG16 and VGG 19 is: VGG 19 adds three more convolution layers before the last three max-pooling layers, which increases the total layers to 19.
AlexNet		A CNN model developed by Krizhevsky et al. [89] in 2012 to compete in the ImageNet competition. AlexNet consists of eight layers with weights. The first five layers are convolutional, and the remaining three are fully connected layers. The output of the last fully connected layer is fed to a 1000-way softmax for the 1000 outcomes' label.
GoogleNet		A CNN model that has 22 convolutional layers, with 27 pooling layers, and nine inception modules under the model linearly [90]. GoogleNet focuses on broader network architecture than deeper.
ResNet		A CNN model that inspires the 34-layer plain network, and the 34-layer residual improves the performance of the 34-layer plain network with insertion of the residual block as shortcut connection [91].

data.

The outcome domain was rated as high risk of bias in 3/7 (43 %) of the studies. All ML studies defined the outcomes in a similar approach for all participants. Only 4/7 (57 %) models used prespecified, standard (Mayo endoscopic score [20]), or consensus-based definitions (UC vs CD diagnosis [27], CD vs non-IBD control [28], and by Crohn's disease activity index [24]) to determine the outcome (signaling Q3.1). In 3/7 (43 %) models the outcome ascertainment was subjective (e.g., inflamed vs non-inflamed), and did not use a previously validated index or criteria.

All seven studies in this category were rated as high risk of bias in the analysis domain. The number of participants with the outcome were considered insufficient (less than 100 participants) in five of 7 studies (71 %) (signaling Q4.1). All studies used imaging data from all enrolled participants (signaling Q4.3). Two of the seven studies provided details on the number of images/frames used from each participant [20,22].

It was not possible to assess the handling of missing data in five studies because no information was provided (signaling Q4.4). Li et al. excluded the imaging data with missing, fuzzy, and incomplete as stated in their exclusion criteria [27], and Maeda Y et al. [20] excluded patients with missing data. To evaluate model performance, one study undertook cross-validation and two studies evaluated their models using an internal validation dataset (signaling Q4.8) [23]. Studies were rated independently in supplementary S3. The overall risk of bias in machine learning studies shown in Fig. 3 and were rated as high risk of bias in 4 domains: participants, outcome, predictors, and analysis domains.

3.3. Overview of image data driven deep learning models

Thirty-three studies utilized DL approaches, all being diagnostic, and 10 studies were prospective. The details of the selected studies are presented in Table 3. Nineteen studies used datasets with more than 100 patients. The largest dataset used 3082 UC patients to develop a model to grade endoscopic disease severity (Mayo endoscopic score) [29]. The majority of the studies either used colonoscopy (n = 19) or wireless capsule endoscopy (n = 7) data. Furthermore, only two studies provided patient race details, with the majority of the training dataset being comprised of White patients (88.3 % and 96 %, respectively) [29,30]. Ten studies provided details of sex distribution, the proportion of male patients ranging from 50 % to 75 % [31–41].

Seventeen studies developed computer-assisted diagnosis (CAD) systems using colonoscopy video images to grade endoscopic disease activity in patients with ulcerative colitis (UC) [11,13,17,19,29] [–] [31,35] [–] [37,42] [–] [48]. Typically, the algorithms predicted either the Mayo Endoscopic Score (MES) (score from 0 to 3) [49] or Ulcerative Colitis Endoscopic Index of Severity score (UCEIS) (score 0 to 8) [50]. Takenaka et al. [30] developed a CAD system that was not only able to predict endoscopic remission (evaluated by UCEIS) from still endoscopic images with an accuracy of 90.1 % (95 %CI, 89.2 %–90.9 %), but also predicted histological remission as defined by Geboes score [51], with an accuracy of 92.9 % (95 %CI, 92.1 %–93.7 %). The inter-rater reliability between the deep neural network algorithm and six experts (3 endoscopists and 3 pathologists) was good-excellent, with kappa coefficients of 0.80 (95 % CI 0.78–0.81) and 0.86 (95 % CI 0.84–0.88) to delineate endoscopic and histological remission respectively. Only one study developed a system using full video data, predicting MES and UCEIS scores with an accuracy of 92.5 % (95 % CI, 88.4 %–96.7 %) and 91.04 % (95 %CI, 86.57 %–95.51 %), respectively [43]. Interestingly, this CAD system [43] incorporated a wireless capsule endoscopy (WCE) feature extraction algorithm [52] into their comprehensive end-to-end automated pipeline. Stidham et al. [29] developed a still image MES classifier, which was incorporated into an automated whole-video MES workflow [53] which performed well with an accuracy of 82.8 % (95 % CI, 71 %–86 %).

Endoscopic video data has the potential of containing imaging artifacts which can create additional noise in model development, and thus filtering these frames prior to model development is imperative. Either a quality control model to automate selective informative frames of interest could be employed, or alternatively non-informative images could be removed by domain experts, prior to model development. Eleven studies incorporated an automated quality control (QC) model in their pipeline. The QC incorporated approaches either focused on excluding images/frames for example with, stool, bleeding related to the procedure (forceps) and water [45], or alternatively included frames with high visual clarity [44].

For CNN based models, Class Activation Maps (CAMs [54]) have been developed to generate visual explanations to localize the important region in the image for prediction. Gradient-weight CAM (Grad-CAM) [55], use class-specific gradient information flowing into the final convolutional layer to produce a coarse localization map. Eight studies undertook post-hoc evaluations, one study generated both CAM and Grad-CAM [11], and seven studies Grad-CAM.

Table 2
Machine learning with medical imaging in inflammatory bowel disease.

Study, Year	IBD Type	Task	Data type	Patients	Model	Number of images for modeling	Cross-validation	Performance
Girgis et al. [22],2010	CD	Detect inflammatory region	WCE	47	SVM	Training: 469 Validation: 253 Test: 277	_	Acc: 0.87
Jebarani et al. [21],2013	CD	Detect inflammatory region	WCE	2	SVM	1828 frames	_	Acc:1.00
Kumar et al. [23],2012	CD	Normal or lesion image (1), lesions classified as mild or severe (2)	WCE	47	SVM	533 images	3-fold cv	Lesion detection: Acc: 0.93 (Whole), 0.91 (ROI), Lesion classification: Acc: 0.94 (Whole), 0.91 (ROI)
Maeda et al. [20], 2022 Pros.	UC	Predict healing or active disease as measured by MES	Colonoscopy	145	SVM	_	_	AI-Active group Acc: (28.4 % [21/74]; 95 %CI, 18.5 %–40.1 %) AI-Healing group Acc: (4.9 % [3/61]; 95 %CI, 1.0 %–13.7 %; P < .001)
Mahapatra D et al. [28], 2013	CD	Discriminate CD from controls	CT	26	RF, SVM, and a Bayesian classifier (BC)	Training: 80 % Test: 20 %	_	Sens: 92.11 % Spec:94.87 % AUROC:0.96
Li T et al. [24], 2022	CD	Discriminate active CD from remission	CT	101	RF, XGBoost, SVM, LR, and KNN	165 images Training: 80 % Test: 20 %	_	Acc: 0.903 Sens: 0.911 Spec: 0.892 AUROC: 0.94
Li H et al. [27], 2019	IBD	Discriminate UC from CD	Multi-slice CT (MSCT)	87	LR, SVM, RF, SGD, LDA	Training: 70 % Test: 30 %	_	Acc: 69.7 % Sens: 80 % Spec: 53.85 % AUROC: 0.81

Pros. is for prospective;
UC: ulcerative colitis, CD: Crohn's disease; IBD: inflammatory bowel disease; CT: computerized tomography (CT);
AUROC: area under the receiver operating curve; Acc: accuracy; Spec: specificity; Sens: sensitivity.

Seven of the studies used images from wireless capsule endoscopy, with the majority (n = 5) evaluating small bowel disease in Crohn's disease. The models either classified and graded CD or defined and detected lesions (mucosal ulcers) [14,56]. At present there is a paucity of histological DL models. Lacucci et al. [57] have developed a new index to assess histological activity in UC, the PICaSSO histologic remission index (PHRI). PHRI evaluates the degree of neutrophil infiltration in the lamina propria and epithelium, the index ranging from 0 to 4. Gui et al. [34] developed a DL cell-classifier tool to detect neutrophils and compute the PICaSSO score automatically. Vande Casteele et al. [39] developed a DL algorithm to detect eosinophilic cells and compute eosinophil density in UC.

Interestingly, Meng et al. [38] explored the use of deep learning algorithms to classify intestinal fibrosis (none, mild, moderate, and severe) in patients with Crohn's disease using CT enterography (CTE) and compared this to a radiomics model (handcrafted features from CTE images). Three-dimensional volumes of interest (VOIs) were annotated by one radiologist as model ground truth and further reviewed by two radiologists. The deep learning model distinguished the degree of fibrosis (normal-mild from moderate-severe) with an AUC of 0.81 (95 % CI: 0.74–0.89) and an accuracy of 0.71.

3.4. Assessment of risk of bias in deep learning models

High risk of bias was observed in the participants domain in the majority of studies (24 of 33). Ten studies used patients from prospective cohorts, five studies utilized data from clinical trials, with four studies utilizing colonoscopy video data; Gottlieb et al. [43] used phase two Mirikizumab trial data, Yao er al [53] used phase two LYC-30937-EC trial data, Schwad et al. [45] utilized UNIFI phase three Ustekinumab trial data, and Gutierrez et al. [44] used datasets from three clinical trials for model development (one phase two randomized, double-blind, placebo-controlled, multicenter rhuMAb Beta7 trial, 2 phase three randomized, double-blind, placebo-controlled Etrozulimab trials) and one clinical trial for testing. Interestingly, Meng J et al. [38] utilized a combination of retrospective and prospective patient data for training and testing respectively, and Lacucci M et al. [37] prospectively recruited patients for model development.

In 10 of the 33 studies, the outcome domain was judged as having a high risk of bias. Twenty-five studies determined the outcome measurement by a pre-specified standard (MES, UCEIS and Geboes score) (Signaling Q3.1). Thirty-one DL studies specified the results in the same manner for all participants, whereas for two studies it was unclear (Signaling Q3.4)[17,18]. Mokter et al. [17] divided their participants into three groups, to derive 3 distinct CNN models: CNN model 1 quantified blood vessels into high and low, CNN model 2 classified images into normal or mild disease, and CNN model 3 classified moderate or severe disease frames.

Eighteen studies were judged as overall high risk of bias for the analysis domain, whereas thirteen studies were unclear, and two studies were deemed low risk of bias. Ten of the 33 studies were high risk of bias that included less than 100 participants, 4 studies were unclear for Signaling Q4.1. Seven studies lacked information on whether all enrolled participants were included in the analysis, 24 studies included data from all enrolled participants (low risk of bias), and two studies selected a subset of patients from all enrolled participants (Signaling Q4.3) [36,38]. Polat et al. [36] excluded a total of 8 patients out of 572 who had images containing debris or artifacts. Meng et al. [37] excluded patients with poor CTE quality. Twenty-seven studies did not provide information on completeness of data and how missingness was handled and thus judged as unclear, while five studies were deemed as low risk of bias [14,30,31,37,38] (Signaling Q4.4). Gui et al. [34] excluded patients with missing data (e.g., missing Mayo endoscopic score, UCEIS-rectum, PICaSSO score) and rated as high risk of bias. Eighteen of 33 studies evaluated model performance using cross-validation, including internal validation (n = 4 utilized on entire dataset, n = 12 additionally

Fig. 3. Risk of bias of AI imaged based studies (N = 33) stratified by model type (ML or DL). Each ML study was analysed by 12 signaling questions, each DL study was analysed by 9 signaling questions. Each domain of probast was analysed separately

undertook cross-validation of the training dataset), or external validation (n = 3). Five studies split the dataset into training, validation, and testing without cross-validation. Takenaka et al. [31] assessed the deep neural ulcerative colitis (DNUC) they developed from a prior study [30] and evaluated the model performance on an external dataset, consisting of 770 patients. Thus, question 4.8 was not applicable to this study. Twenty-four studies were ranked as low risk of bias (Signaling Q4.8). The overall risk of bias in deep learning studies was rated as: high risk of bias in 3 domains, participants, outcome, and analysis domains.

4. Discussion

4.1. Study bias overview

In this study we assessed the methodological quality of studies developing machine learning or deep learning image based IBD prediction models. Overall, all studies exhibited high risk of bias in all domains (ML studies were assessed in 4 domains and DL studies assessed in 3 domains). The analysis domain was commonly rated as high risk of bias for both types of studies, because of unclear information provided on whether all enrolled participants were included in the analyses (Q4.4). Furthermore, the majority of the ML studies used a low number of participants with the outcome relative to predictors (Q4.1) for model development, varying from two to 145 participants. Deep Learning studies were larger, participant size ranging from two to 3082, and 10 studies were prospective, and with one study enrolling participants specifically for model development.

4.2. Implications of data quality

All the ML models were developed using data from a single center, and only 12 (36 %) DL studies used multicenter data. The development of a model in the pre-clinical setting should entail data representative of the population, capturing the disease heterogeneity, and which most likely entails large multicenter datasets. Furthermore, developing models using data from various sites, will capture potential technical differences and variations in clinical practice, improving the generalizability of a model for implementation in the clinical setting [58]. It is common to undertake secondary data analysis, using pre-existing data for model development. Nine (27 %) studies used clinical trial data, and although these provided large datasets for model development, a consideration is whether the data is representative of the "real-world", i.e., encompassing the complexities of the healthcare system and the heterogeneity of the disease. The optimal sample size for training ML/DL models should be sufficient to capture the underlying variations and patterns in the data, that encompasses the complexity of the problem and dimensionality of the data. Furthermore, selection bias can arise from missingness, specifically if non-random. The ultimate goal is to incorporate these algorithms into the clinical workflow, to improve both efficiency and patient care. Of all the studies, only one used a publicly available source dataset [15]. Understanding the source data is imperative, a common pitfall is using poor quality data, hampering reproducibility. One, such example is the development of duplicated datasets, "Frankenstein data" [59]. These datasets are compiled from different datasets and distributed under a new name. Merging different datasets has potential implications, firstly if the labelling convention is not the same, data could be mis-labeled, datasets have their own biases, and combing could create new biases and combining data from different sources could result in a domain shift and thus have poor generalizability.

Table 3
Deep learning with medical imaging in inflammatory bowel disease.

Study, Year	IBD Type	Task	Data type	Patients	Model	Number of images for modeling	Cross-validation	Performance
Ozawa T et al. [42], 2019	UC	Distinguish MES 0 and 0-1	Colonoscopy	955	GoogLeNet with 22 layers	Training: 26304 frames Validation: 3981 frames	–	AUROC: 0.86 AUROC: 0.98
Stidham et al. [29], 2019	UC	Predict MES	Colonoscopy	3082	Inception-v3 with pre-trained on ImageNet	Training: 16514 images (Mayo 0:8951, Mayo 1: 3584, Mayo 2:2278, Mayo 3:1701) 80 % images; Validation: 10 %; Test:10 %	10-fold cv	AUROC: 0.97
Takenaka et al. [30], 2020 Pros.[a]	UC	Endoscopic remission, Histologic remission, UCEIS score	Colonoscopy	2887	Inception-v3	Model development: 40758 images, 6885 biopsies Model evaluation: 4187 images, 4104 biopsies	–	Acc: 0.93
Mokter et al. [17],2020	UC	Classify UC, MES	Colonoscopy	N/A	Inception-v3 159 layer	178 video clips, 67599 frames, 4295444 patches		Acc: 0.78
Gutierrez et al. [44], 2021[a]	UC	Quality control model, and predict MES	Colonoscopy	1105	ResNet50 pre-trained on ImageNet	Quality Control (QC) model: 4371 frames (Training: 70 %, Validation: 20 %, Test: 10 %) Ulcerative Colitis Scoring (UCS) model evaluation: 1672 videos	5-fold cv	AUROC (Mayo 1): 0.84 AUROC (Mayo 2): 0.85 AUROC (Mayo 3): 0.85
Turan et al. [12],2022	UC	UC severity classification, Ulcerative Colitis Endoscopic Index of Severity (UCEIS)	Colonoscopy	N/A	ConViT, Inception-v4, NfNets, ResNet & Swin Transformer, pre-trained on ImageNet	673 images Each fold: 134 images	5-fold cv	Acc: 0.85
Sutton et al. [11],2022	UC	Discriminate UC from non-UC, Grading UC	Colonoscopy	1229	Inception-v3, ResNet 50, VGG19, DenseNet121, pre-trained on ImageNet	Training: 80 % with 5-fold cv Test: 20 %	5-fold cv	Acc: 0.88 AUROC: 0.90
Takenaka et al. [31], 2022 Pros.[a,b]	UC	Detect histological mucosal inflammation, UCEIS	Colonoscopy	853	DNUC from Takenaka et al. [26]	External validation based on prior study	–	Acc: 0.95 Spec: 0.95 Sens: 0.98
Lo B et al. [19],2022	UC	UC severity classification, MES	Colonoscopy	467	Inception-v3, EfficientNet	Training: 85 % with 5-fold cv Test: 15 %	5-fold cv	Acc: 0.84
Polat G et al. [36],2022	UC	UC severity classification, MES	Colonoscopy	564	ResNet18, ResNet50, DenseNet121, Inception-v3, MobileNet-v3-large, pre-trained on ImageNet	Training: 85 % with 10-fold cv from 462 patients Test: 15 % from 85 patients	10-fold cv	Acc: 0.772 Sens: 0.693 Spec: 0.911
Luo et al. [13],2022	UC	Endoscopic remission, MES	Colonoscopy	1317	VGG19, ResNet152, Inception-v3, DenseNet201, UC-DenseNet (DenseNet201 + IndRnn + EAM-Net)	Training: 60 % Validation: 20 % Test: 20 %	–	Endoscopic remission: Average Acc: 0.9825 Average AUC: 0.9815 Mayo classification: Average Acc: 0.911
Huang TY et al. [48], 2021	UC	Endoscopic mucosal healing, MES	Colonoscopy	54	DNN, KNN, SVM, Combined model	Training: 70 % Test: 30 %		Acc: 94.5 % Sens: 89.2 % Spec: 96.3 %
Ruan et al. [92],2022 Pros.[a,b]	UC/CD	Discriminate UC from CD	Colonoscopy	1772	ResNet50, pre-trained on ImageNet	49154 images Training: 47322 images from 1576 patients External validation: 1832 images from 196 patients	6-fold cv	Acc (patient level): 0.96 Acc (lesion level): 0.92
Wang et al. [32],2022	UC/CD	Distinguish UC/CD/control	Colonoscopy	596	ResNet-50, ResNet-101, ResNeXt-50, ResNeXt-101, EfficientNet-V2. ResNeXt-101, pre-trained on ImageNet	15330 images Training: 13872 images (Validation: 10 %) Testing: 1458	–	Acc (image level): 0.92 Acc (patient level): 0.91
Chierici M et al. [93], 2022	IBD	Classify UC and CD, HC and IBD	Colonoscopy	N/A	ResNet18, ResNet34, ResNet50, ResNet101, and	Training: 70 % Validation: 20 % Testing: 10 %	5-fold cv	MCC >0.9 on HC and IBD

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Table 3 (*continued*)

Study, Year	IBD Type	Task	Data type	Patients	Model	Number of images for modeling	Cross-validation	Performance
					ResNet152, pre-trained on ImageNet			MCC >0.6 on CD and UC
Yamamoto et al. [40], 2022[a]	IBD	Classifying neoplasia occurring in IBD	Colonoscopy	63	EfficientNet-B3 after Cycle_GAN, pre-trained on ImageNet	Training: 80 % patients with 3-fold cv Test: 20 %	3-fold cv	Sens:72.5 % (95 % CI:60.4–82.5) Spec: 82.9 % (95 %CI:74.8–89.2) Acc:79.0 % (95 % CI:72.5–84.6)
Yao et al. [53],2021 Pros.[a,b]	UC	Informative frame detection	Colonoscopy (video)	157	Inception-v3 with pretrained ImageNet weights	Training: Developed on internal: 30 of 51 videos (34810 frames) Testing on external multicenter clinical trial: 10 of 264 videos	5-fold cv	Acc: 0.844 AUROC: 0.93
Schwad et al. [45], 2021 Pros.	UC	Predicting mucosal healing (1) and remission of UC(2)	Colonoscopy (video)	726	SVM, ResNet, pre-trained on ImageNet	1881 videos (in a total of 1,129,188 frames)	5-fold cv	AUROC: 0.92 AUROC: 0.90
Gottlieb et al. [43], 2021 Pros.[a]	UC	Predict MES and UCEIS score	Colonoscopy video	249	Bidirectional RNN	795 (411 videos with cecum, 384 videos without) Training: 80 % with 5-fold cv Testing: 20 %	5-fold cv	Ems Acc: 0.93 UCEIS Acc:0.91
Klang et al. [94],2021	UC	Distinguish between strictures and normal mucosa, UC severity score	WCE	Klang et al. [29] Barash et al. [38]	EfficientNet-B5	27892 images 10 patient-level experiments (1 patient per fold)	Patient-level cv	AUROC (strictures vs. Mild, moderate, and severe ulcers): 0.99, 0.98, and 0.89,
Higuchi et al. [46], 2022	UC	Severity of UC, MES	WCE	22	ResNet50	31 WCE videos Training: 483644 images Testing: 255377 images	–	Acc: 0.97
Klang et al. [56],2020	CD	Detect mucosal ulcers	WCE	49	Xception, pre-trained on ImageNet	Training: 80 %, Testing: 20 % Individual patient level, n-1 patients for training, 1 for testing, and repeat 10 times	5-fold cv	Acc: 90.5-0.97 AUROC: 0.99
Klang et al. [95],2021	CD	Detecting images containing NSAIDs mucosal ulcers	WCE	124	Xception ImageNet pretrained	Training: 17640 CD images (7391 mucosal ulcers, 10249 normal mucosa: 3577 normal CE + 6672 CD), Testing: 1605 NSAID images (980 with NSAID ulcers, 625 normal mucosa)	–	NSAID AUROC: 0.97
Barash et al. [47],2021	CD	Classify and grade CD, Pillcam CD classification	WCE	49	CNN, 6 ResBlocks of 3 layers each (designed by study)	Training: 1242 images (500 mild, 304 moderate, 438 severe) Testing: 248 images	5-fold cv	Acc: 0.67
Majtner et al. [96], 2021 Pros.[a]	CD	CD detection, Lesion severity classification	WCE	38	ResNet50, pretrained on ImageNet	7744 images Training: 70 % Validation: 10 % Testing: 20 %	–	Acc: 0.98
De Maissin et al. [14], 2021[a]	CD	Define quality of 8 types lesions annotation	WCE	63	ResNet34, VGG16, VGG19, pretrained on ImageNet	66 videos Round 1: 3498 images Round 2: 1641 images Round 3: 3498 images	5-fold cv	Acc (resnet34): 0.95
Lamash et al. [18], 2019	CD	Quantitative assessment of CD from MRI	MRI	23	U-net	Training: 15 cases Test: 8 cases	–	DSCs of 75 ± 18 %, 81 ± 8 %, and 97 ± 2 % for the lumen, wall, and background,
Gui et al. [34],2022 Pros.[a]	UC	Define histologic remission index (PHRI), Identify active from quiescent UC	Pathology	307	VGG16	614 biopsies Training: 70 % (15 % validation) Testing: 30 %	–	Acc: 0.86
Casteele et al. [39], 2021	UC	Eosinophil density and clinicopathologic features	Pathology	88	CNN	Training: 3480 eosinophils	–	Interclass correlation coefficients: 0.81–0.92
Ohara et al. [35],2022	UC	Predict endoscopic remission by	Biopsy	114	Feature Pyramid Network (FPN)	Training set 1: Training 735; Validation 165	–	Detected goblet cell mucus: Acc: 0.97

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Table 3 (*continued*)

Study, Year	IBD Type	Task	Data type	Patients	Model	Number of images for modeling	Cross-validation	Performance
		quantification epithelial cell and goblet cell, MES				Training set 2: Training 300; Validation 83		Sens: 0.83 Spec: 0.98
Mizuno et al. [41], 2022	IBD	Predict the development of pouchitis (binary) after IPAA in UC patients.	CE (endoscopic pouch images)	43	3 layers CNN	5-fold cv with no Test set	5-fold cv	AUROC: 0.84
Lacucci M et al. [37], 2022 Pros.[a]	UC	Endoscopic remission: UCEIS Histologic remission: Paddington International virtual chromoendoscopy score (Picasso). RHI, NHI, PHRI	Virtual chromoendoscopy (VCE) and white-light endoscopy (WLE)	283	ResNet-50	WLE - UCEIS: Train: 239 videos, Validation 58 videos, Test 242 videos VCE - PICaSSO: Train: 245 videos Validation 66 videos Test 244 videos	–	Distinct ER (primary outcome): Sens: 79 % (95 % CI:63 %–90 %), Spec:95 % (95 % CI 91 %–98 %), AUROC:0.94 (95 %:CI 0.91–0.97) Acc: 92 % (95 % CI: 88 %–95 %)
Meng J et al. [38],2022 Pros.[a]	CD	Distinguished none-mild from moderate-severe bowel fibrosis	CT Enterography	235	3D deep convolutional neural network based on ResNet	Training: cohort 1: 114 patients with 159 segments Test: 65 patients with 97 segments (center 1) 35 patients with 35 segments (center 2) 21 patients with 21 segments (center 3)	10-fold cv	Average AUROC: 0.82 Average Acc: 0.72 Average Sens: 0.70 Average Spec: 0.76

Pros. is for prospective;
WCE: wireless capsule endoscopic;
UC: ulcerative colitis, CD: Crohn's disease; IBD: inflammatory bowel disease; MES: Mayo endoscopic severity; IPAA: ileal pouchanal anastomosis; UCEIS: Ulcerative Colitis Endoscopic Index of Severity;
AUROC: area under the receiver operating curve; Acc: accuracy; Spec: specificity; Sens: sensitivity; MCC: Matthews correlation coefficient; DSC: Dice similarity coefficient.
[a] is for multicenter study;
[b] External validation.

4.3. Algorithmic bias concerns

Only a minority of studies considered in this systematic review provided participant race and sex distribution information for model development. Algorithms need to perform equally across a diverse heterogenous population, with the burden of IBD rising globally [60]. Moreover, algorithmic bias is of concern, as the majority of the models were trained on historical data, and the potential for racial and sex biases, that maybe incidentally encoded [61]. Societal and racial discrimination maybe amplified, and with models performing disproportionately worse in under-represented populations [62,63]. All models should be evaluated across a diverse population prior to deployment in a clinical setting.

4.4. Clinical implications

The majority of DL studies either aimed to predict endoscopic disease indices from still endoscopic images or characterizing lesions (e.g., ulcers) from wireless capsule endoscopy images. The training data sets, labeled with the outcome measure are dependent on expert knowledge. At present there is no clear guidance on the minimum number of annotators required to label each image, to account for potential variation and biases in assessment. The Checklist for Artificial Intelligence in Medical Imaging (CLAIM), which was primarily developed for radiological AI studies, advises reporting on the specific number of human annotators and training provided and their qualifications [64]. Authors should provide details on whether annotations were done independently and how discrepancies were resolved. Takenaka et al. employed three pathologists to ascertain the ground truth label and one pathologist to resolve discrepancies [30]. Moreover, studies need to choose suitable measurements to evaluate the concordance between the ground truth and model results [65].

At present, the treatment goal in IBD is to achieve endoscopic mucosal healing [66], and possibly over time it may evolve to include histological remission [67] in UC or transmural healing in CD [68]. The focus of the majority of DL models was to recapitulate the UC endoscopic scoring indices. There is a paucity of AI models to assess endoscopic disease severity in Crohn's disease, which is inherently more challenging due to its "patchy" distribution. The current indices, which are subject to human interpretation may become obsolete as therapeutic goals change, and efforts maybe best directed at developing quality datasets for segmenting and characterizing specific endoscopic features, such as cobblestoning, deep ulcerations etc., for future index development [69,70].

4.5. Evaluating model performance and generalizability

Typically, model performance is evaluated in two steps, internal validation followed by external validation. We found 1/7 ML and 18/33 DL studies evaluated models using cross-validation (typically 5-fold [71]), testing a models ability to predict new data and to identify overfitting. A generalizable model is required prior to implementation in practice, and it is closely related to both representative input data and model validation [72]. Prospectively validating a model using an external dataset is important to assess the stability (generalizability) of the model in different clinical settings [73]. Only three studies used prospective external validation cohorts to test. Takenaka et al. [31] independently validated their UC endoscopic and histological remission CNN model [30] on an external cohort. Yao et al. [53] developed an Inception-V3 deep learning model using their local institution data, and evaluated the model on multicenter clinical trial. We noted variability in the performance matrices used to evaluate the models, standardization would need to be addressed in future guidelines.

4.6. Future directions

The models evaluated in this systematic review were diagnostic in nature. The ultimate goal of such models is to be embedded into a computer aided diagnosis (CAD) system for automated decision making in IBD. CAD systems for IBD are in their infancy, with the majority using still images rather than videos suitable for real-time evaluation and application. Comparatively, notable advances have been made in developing CAD systems to detect colonic polyps. Six prospective randomized controlled trials (5 Chinese, 1 Italian) using in vivo CAD real-time systems have been evaluated between 2019 and 2020 [74–79]. At present, three polyp detection CAD systems have regulatory (FDA) approval [80]. To achieve comparable success, in our opinion, we suggest collaborating to create large, high-quality, and diverse IBD imaging datasets.

Efforts need to be undertaken to standardize and improve design quality, to advance the adoption and deployment of these algorithms in the clinical setting. At present no clear reporting requirements have been developed to serve as a guidance. Currently, under development is the STARD-AI [81] guidelines to assess diagnostic test accuracy and performance and TRIPOD-AI [82] focusing on evaluating multivariable AI prediction models, which could serve as aids to improve the current standards.

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CRediT authorship contribution statement

Xiaoxuan Liu: Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **James Reigle:** Data curation, Writing – review & editing. **V.B. Surya Prasath:** Methodology, Supervision, Writing – review & editing. **Jasbir Dhaliwal:** Conceptualization, Data curation, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

Xiaoxuan Liu MS, James Reigle, Surya Prasath PhD and Jasbir Dhaliwal have no declaration of interests.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compbiomed.2024.108093.

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